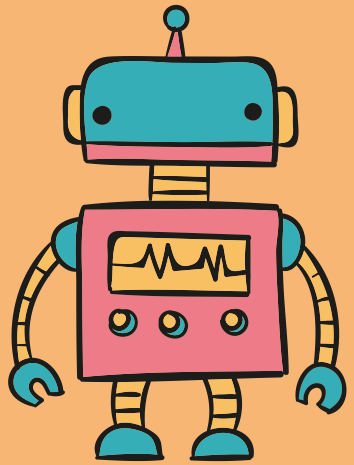
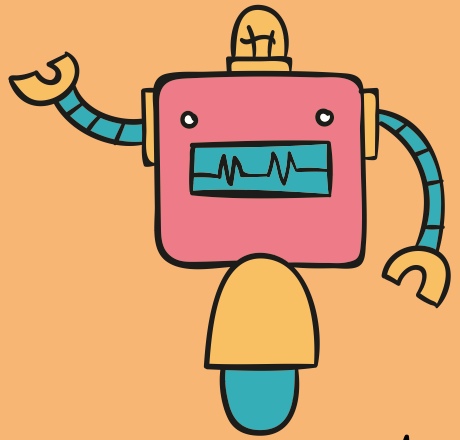
AI Medical Chatbot





By: Group 4

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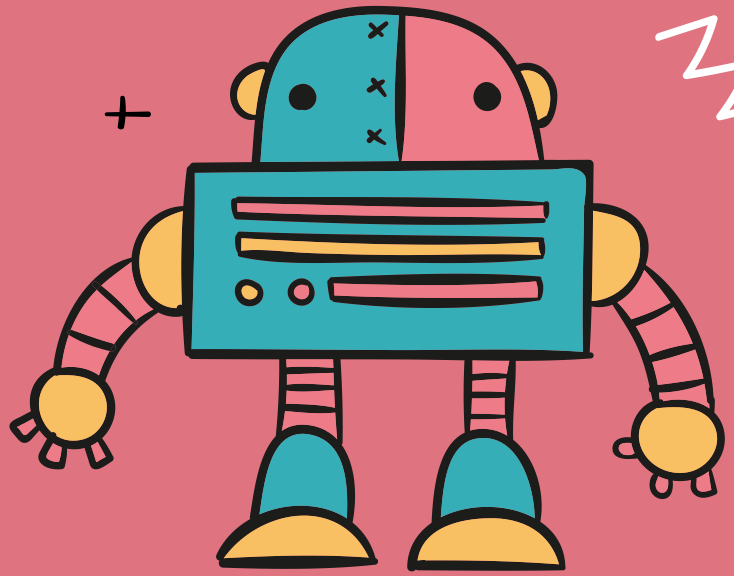

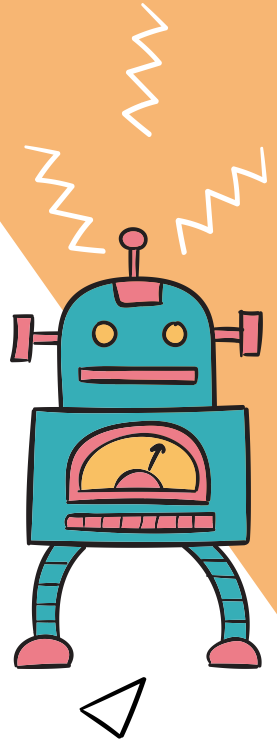

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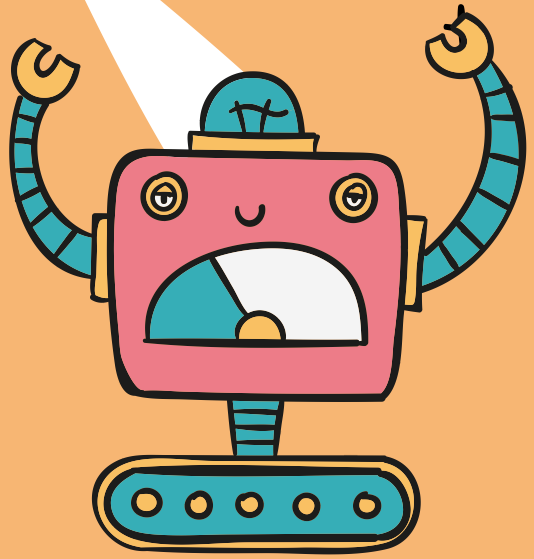
01.

Motivation and Overview



Background

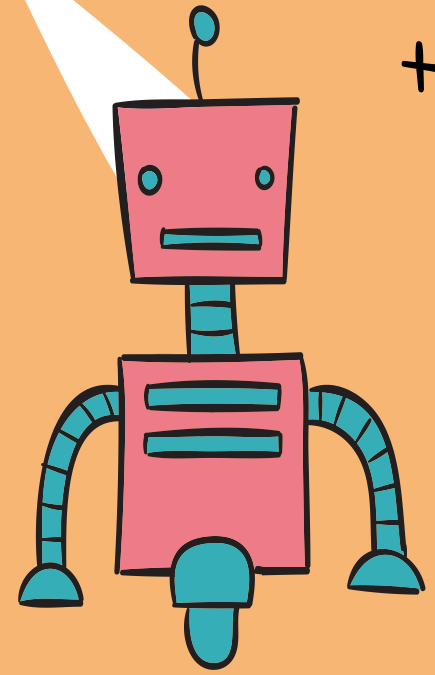
- Access to reliable medical information is a recurring medical need in society.
- Modern large language models (LLMs) now provide responses to free-form medical symptom questions.
- **Yun & Bickmore (2025)** revealed that about 29% of people now use LLM chatbots to search for medical information, out of which 90% of respondents used online search or chatbots to look up health conditions

Solution Proposed

- Fine-tune a small transformer model (FLAN-T5 base) on 257k medical QA pairs using LoRA due to computational constraints
- Augment with real-time retrieval (RAG) from authoritative medical sources
- Perform comprehensive evaluation using ROUGE-L score, BERTScore-F1
- Compare results using 3 variants
  - Base T5
  - Base T5 + LoRA Fine-tuning
  - Base T5 + LoRA Fine-tuning + RAG

02.

Data Preprocessing



Dataset Description

- The raw dataset is retrieved from HuggingFace: https://huggingface.co/datasets/ruslanmv/ai-medical-chatbot/

- Dataset size: (256916, 3)

- The columns of the dataset is as follows:
  - Description: Original question
  - Patient: Context (Background) of the patient
  - Doctor: The doctor's reply

Description string · lengths 1↔152 99.8%	Patient string · lengths 1↔1.78k 99.3%	Doctor string · lengths 2↔1.14k 95.1%
Q. What does abutment of the nerve root mean?	Hi doctor,I am just wondering what is abutting and abutment of the nerve root means in a back issue. Please explain. What treatment is required for annular bulging and tear?	Hi. I have gone through your query with diligence and would like you to know that I am here to help you. For further information consult a neurologist online -->

Data Preprocessing

- Data preprocessing is done offline and results were saved as CSV to be utilised

- **Data Cleaning include:**
  - Deduplication of records
  - Non-ASCII removal using Regex
  - Whitespace normalisation

- **Length Filtering include:**
  - Minimum Length Threshold to remove short incomplete entries (min_desc_length = 10, min_doctor_length = 20)
  - Maximum Length Threshold to prevent truncation during training (max_length = 1000)

Conversational Text Processing

- Remove some redundant text strings to improve output results.

- Using a ChatBot, users will ask:

  "What are symptoms of diabetes"

  instead of

  "Hi doctor, what are symptoms of diabetes, thank you."

- Text strings to be removed include:
  - Greetings ('Hi Doctor', 'Hello Doctor')
  - Appreciation ('Thank you', 'Many thanks')
  - Signature ('Best regards, Dr. Sam', 'Wishes, Dr. John')

Format Construction and Split

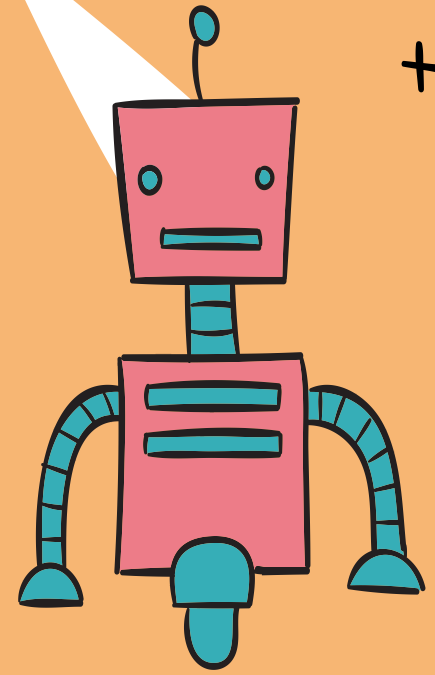
- The input is formatted into this structure to work well with T5:
  - **Formatted_Input:**
    'Answer this medical question:
    Medical Question: [question]
    Patient Information: [context]'
  - **Answer:** [doctor response]

- Train/Val/Test Split: 80/10/10

03.

Model Architecture

with LoRA



Why FLAN-T5-Base?

- **FLAN (Fine-tuned Language Net)**
  - Already pre-trained on instruction-following task
  - Already understands 'Answer this question:' prompt (just like how we structure out QA pairs to be)

- **T5 (Text-to-Text Transfer Transformer)**
  - Encoder-decoder architecture
  - Good for QA pairs language tasks

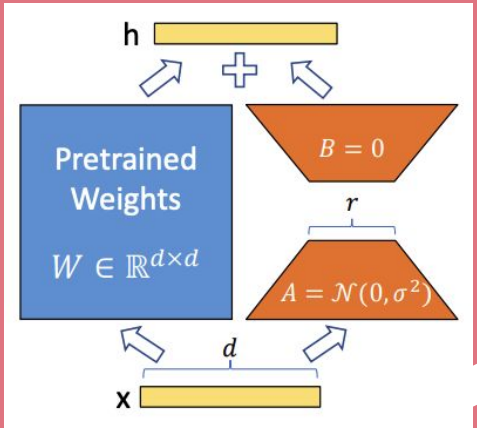
- Model Description
  - 220M parameters
  - 12 encoder + 12 decoder layer
  - Hidden dimensions: 768

LoRA Fine-Tuning

- It is nearly impossible to fine-tune FLAN-T5 base since the process of update all parameters (220 million parameters) is computationally expensive

- **Low-Rank Adaptation (LoRA)** is employed to freeze the base model weights and inject small trainable adapter matrices

- We used the following LoRA configurations:
  - rank = 32 (size of low rank decomposition)
  - lora_alpha = 32 (scaling factor)
  - lora_dropout = 0.1
  - bias = none



LoRA Parameters Breakdown

Since hidden size = 768,

With LoRA (Rank =32),

number of trainable parameters per weight matrix = 768 x 32 + 32 x 768 = 49,152

Number of layers = 12

Components: Encoder Self-Attention, Decoder Self-Attention, Decoder Cross-Attention

Target matrices = q, k, v, o

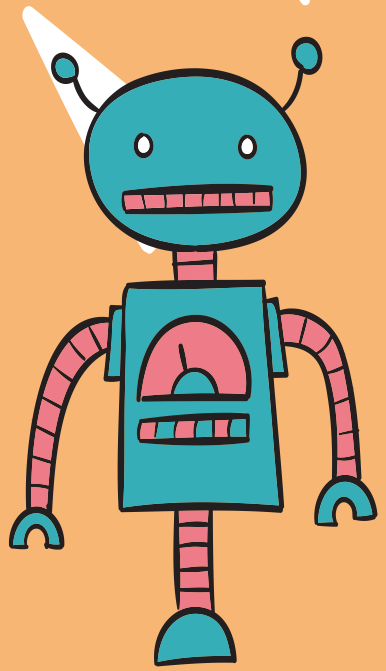
Total number of parameters = 12 x 3 x 4 x 49,152 = 7,077,888 (7.08M)

Comparing to base model without LoRA,

It is about **3%** of the total parameters that need to be updated

04.

Training Pipeline

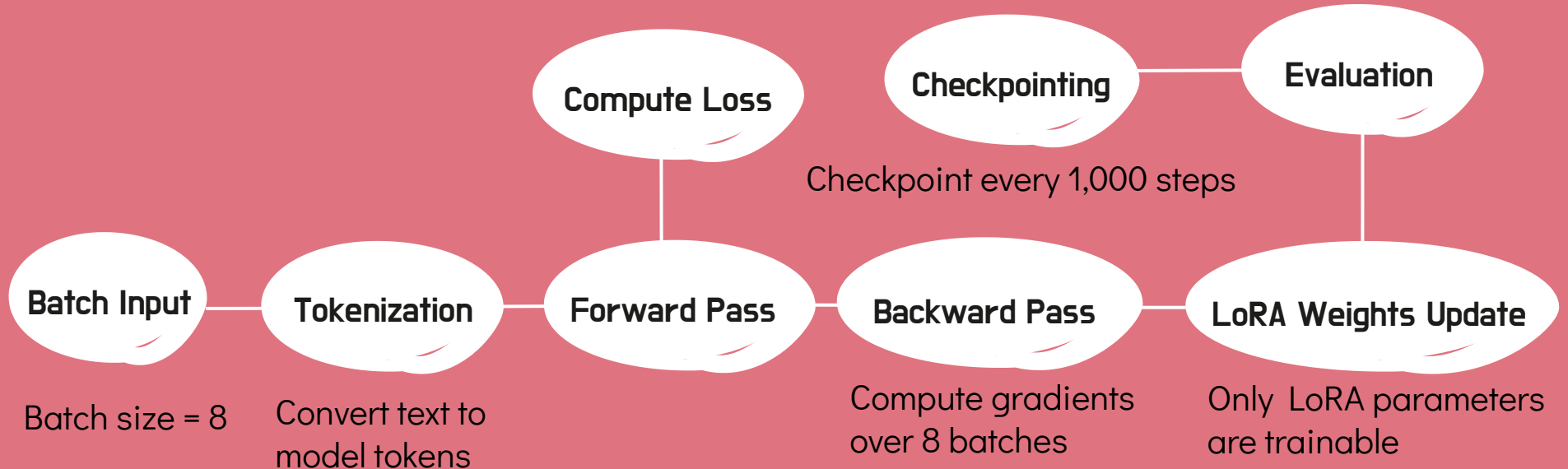


Training Setup

- **Tokenization**
  - Start by tokenizing text into token IDs that model can process while limiting the maximum tokens to be 512 (truncation of too long)
  - **Tokenizer used: FLAN-T5's pre-trained tokenizer (32,000 tokens vocab)**

- Training hyperparameters
  - Learning Rate: 5e-4
  - Batch size: 8 per device
  - Gradient Accumulation: 4 steps (Effective batch size = 8 x 4 = 32)
  - Epoch = 2 (LoRA converges quickly)
  - Optimizer: **Adafactor**
    - Memory-efficient and well-suited for T5 fine tuning

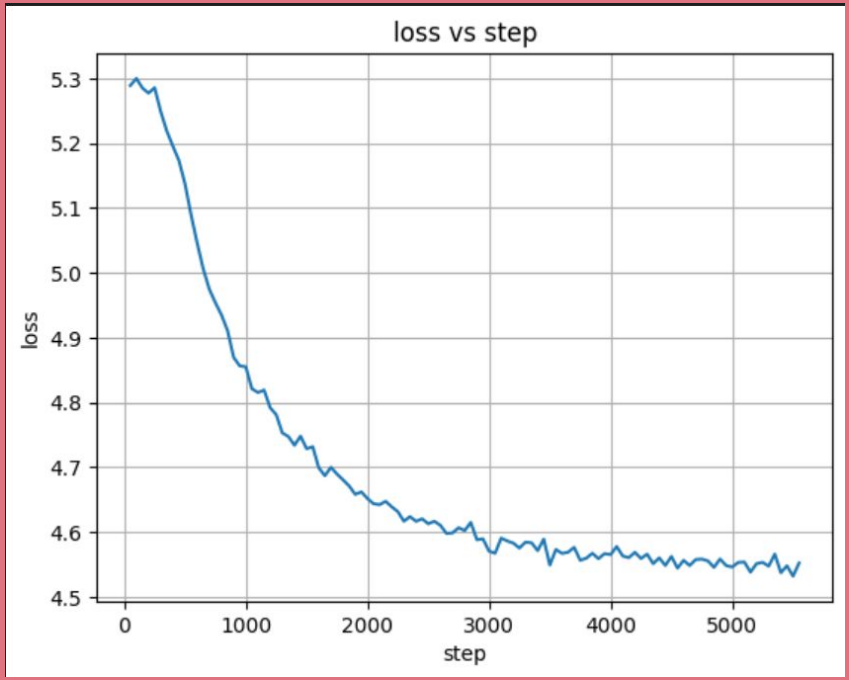
Overall Pipeline



How we selected the best model

- Evaluation was performed after each checkpoint on a validation dataset, tracking loss and qualitative outputs.
- The best-performing checkpoint was chosen for final testing.
- The validation-loss plot is shown on the next slide

Training Convergence



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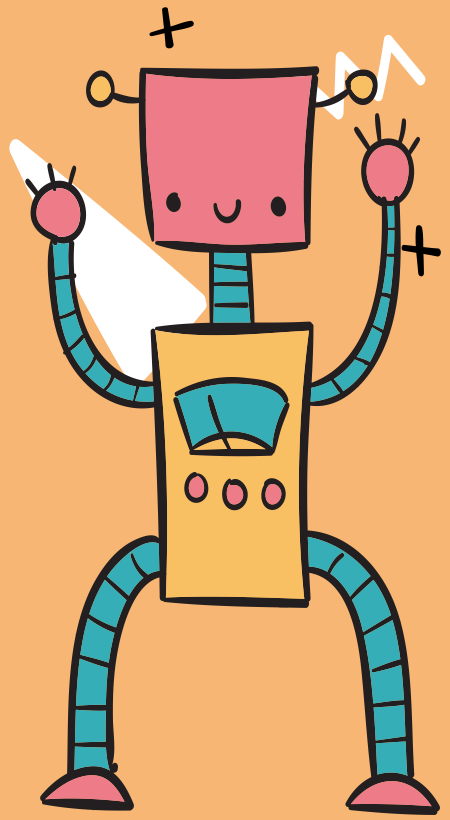
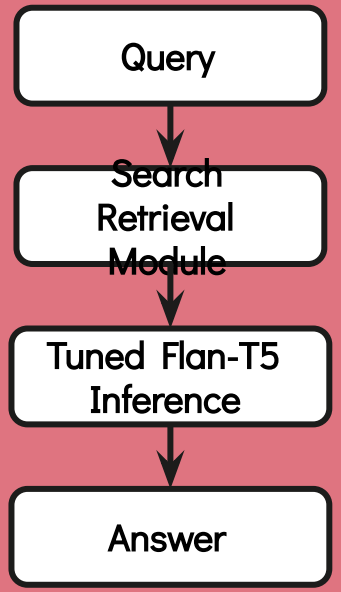
RAG with Beautiful Soup



Grounding with Facts

- **Retrieval Augmented Grounding (RAG)** is employed to enhance model responses with real-time information retrieval from to web as 'ground truth'
- Specifically, we grounded the answers in authoritative medical sources
- Sources were retrieved through **Google Custom Search API** and scraped using **BeautifulSoup** to get clean text
- The information retrieved were appended to user's query to provide model with more external knowledge other than dataset to produce more accurate output

System Workflow Overview



Formatted with prefixes same as training task
"Answer this medical question, Question: ... Patient Information: ..."

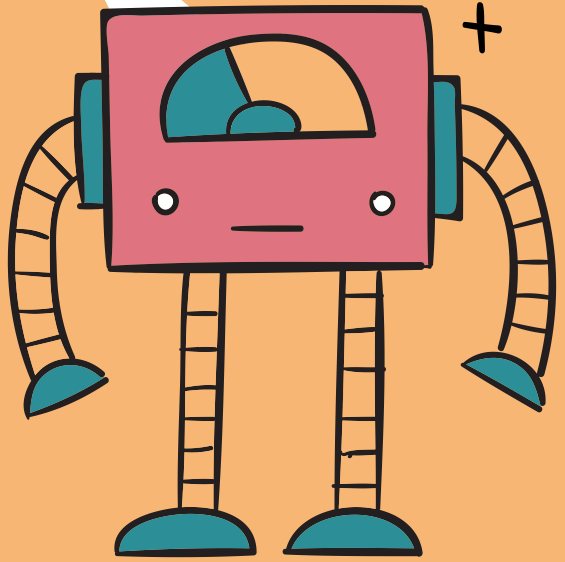
Pings Google Search API with query, **returns top 1 result & reference**
Result appended to the Query before being sent for inference

Adapter weights added to model, inference conducted with beams=4

Answer returned to user with reference

06.

Results and Analysis of Methods



Quantitative Results

- Final Eval Loss: **~4.696**
- Quantitative comparison
- We used the following evaluation metrics:
  - ROUGE-L (measures textual overlap)
  - BERTScore (measures semantic similarity)

	T5-base	T5-base + LoRA	T5-base + LoRA + RAG
ROUGE-L	0.02	0.08	0.15
BERTScore-F1	0.79	0.80	0.85

Qualitative Results

```
================= Base Model =================

Q: can i get pregnant if my sonography shows bulky ovaries with small follicles?
P: recently i had a sonography test where the report says that my both ovaries appear bulky. left ovary measures 12cc and right ovary measur
es 7cc and multiple small follicles are seen in both ovaries. my question to you is whether i can be pregnant or not . my age is 26yrs old
A: my sonography test says that my both ovaries appear bulky. left ovary measures 12cc and right ovaary measures 7cc , and multiple small fo
llicles are seen in both ovaries. my question to you is whether i can be pregnant or not . my age is 26yrs old
================= Fine Tuned Model =================

Q: can i get pregnant if my sonography shows bulky ovaries with small follicles?
P: recently i had a sonography test where the report says that my both ovaries appear bulky. left ovary measures 12cc and right ovary measur
es 7cc and multiple small follicles are seen in both ovaries. my question to you is whether i can be pregnant or not . my age is 26yrs old
A: and welcome to hcm. i have read your query and i will be happy to help you. if your ovaries are bulky, then you may be pregnant. you may
have a uterus or ovary, if you have ovarian abnormalities, then i would suggest you to consult a fertility specialist. hope i has answered y
our query. let me know if i can assist you further.
```

```
================= Final Answer (RAG) =================

Q: can i get pregnant if my sonography shows bulky ovaries with small follicles?
P: recently i had a sonography test where the report says that my both ovaries appear bulky. left ovary measures 12cc and right ovary measur
es 7cc and multiple small follicles are seen in both ovaries. my question to you is whether i can be pregnant or not . my age is 26yrs old
A: thanks for your query. i understand your concern. if your ovaries are bulky, then you may have a polycystic ovary syndrome, which is a co
ndition in which the ovarian follicles are in the ovulation and menstrual periods. ovules are a common cause of ovary syndrome. so, if you h
ave ovarious ovaries, you may be able to get pregnant if i can help you. hope i have answered your question. let me know if I can assist you
further.
Sources:
 [1] Polycystic ovary syndrome: MedlinePlus Genetics - https://medlineplus.gov/genetics/condition/polycystic-ovary-syndrome/
```

Analysis of Methods - Fine Tuning

Pros	Cons
Conservative Learning Rate + use of Adafactor (static config) - stability during training	**May not be as optimal** - less learning. For the model used in demo (v5) - disabling Adafactor's natural adaptive step scaling may have resulted under-utilised learning capacity
LoRA only on attention - for lower memory usage - fewer weights + stable baseline	**Seems to do little for domain knowledge injection** , merely style. Tuning additional components like FFN layers would better modify the model's internal mapping for medical relationships, at the cost of more trainable parameters + longer training time
Label smoothing - to reduce overfitting	
2 epochs - due to faster expected convergence, training therefore can run faster	

Analysis of Methods - RAG

Pros	Cons
Improves Factual Accuracy - RAG grounds model responses in real, external data from **trusted domains** (CDA, MOH, CDC, MedlinePlus).	**Latency and Complexity** - Each query performs external searches and web scraping, increasing **latency vs. pure generation** .
Freshness and Adaptability - the retriever dynamically pulls **up-to-date** medical information without retraining the model.	**Dependence on Retriever Quality** - If retrieval fails (e.g., API rate limits, poor keyword matching), generation quality drops sharply.
Reliability - answers include retrieved context + citations, allowing users (and evaluators) to verify facts.	**Domain Bias** - Even whitelisted sources can contain outdated or country-specific guidelines (e.g., CDC vs. MOH)
Better Domain Specialization - Retrieval can be scoped to whitelisted medical websites, giving **domain precision without** ***overfitting***	

Next Steps

- Low ROUGE-L score is likely due to **underoptimisation**
  - Adafactor able to run with natural adaptive step scaling - trying this next
  - Currently, only attention weights tuned - could extend this to the FFN

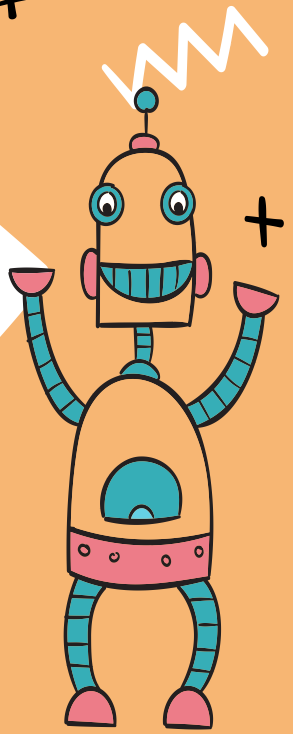
- **Final 1-2 runs of train** -> evaluation before report submission
  - Code mostly templated, matter of switching around hyperparameters
  - Aim is to increase ROUGE-L Score
  - Main comparison: Dynamic vs static step scaling (1 run)
  - Secondary comparison: Adding in tuning for the FFN

- However - likely limited dataset (180K rows) doesn't fully encompass the domain of medical knowledge - **limited gains expected**

07.

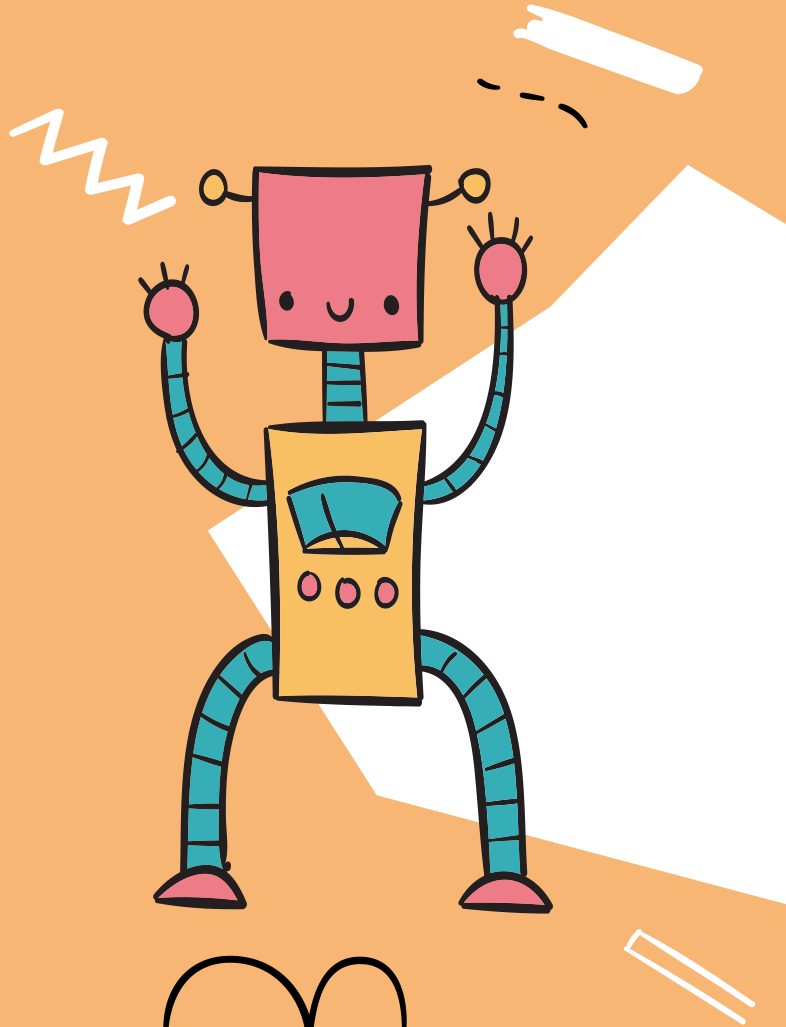
Live

Demo



Thanks!

Any QUESTIONS?



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References

Yun, H. S., & Bickmore, T. (2025). Online Health Information Seeking in the Era of Large Language Models: Cross-Sectional Online Survey Study (Preprint). *Journal of Medical Internet Research*. https://doi.org/10.2196/68560

https://arxiv.org/pdf/2106.09685